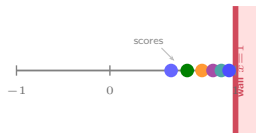


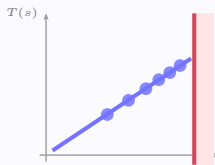
Cosine similarities

(bounded in $[-1, 1]$)



$$T(x) = x$$

Gumbel coordinate: $T(x) = x$ (InfoNCE)



Scores **compressed**
near the boundary.

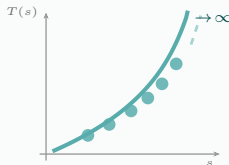
Tail structure is
non-additive.



wall persists

Weibull coordinate: $T(x) = -\log(1 - x)$ (WEINCE)

$$T(x) = -\log(1 - x)$$



Boundary "unfolded"
to $+\infty$. Scores
spread out.

$T(X) = \eta(s) + \varepsilon$
is now **additive**.



wall unfolded

$T_{\text{Gumbel}}(x) = x$ and $T_{\text{Weibull}}(x) = -\log(1 - x)$ are **non-affinely related**.

$\Rightarrow$ No temperature rescaling x/τ can convert one into the other.